Analyzing Zigbee Traffic: Datasets, Classification and Storage Trade-offs

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Abstract

Zigbee is widely used in smart home environments due to its low power consumption and support for mesh networking, making it a relevant target for traffic-based IoT forensic analysis. However, existing studies often rely on limited datasets and fixed network configurations. In this paper, we analyze Zigbee network traffic from three complementary perspectives: data collection, traffic classification, and storage efficiency. We introduce ZIOTP2026, a publicly available dataset of Zigbee traffic collected from commercial smart home devices deployed under multiple network configurations and capturing realistic interaction scenarios. Using this dataset, we study two traffic classification tasks: device type classification and individual device identification, and evaluate their robustness under both intra-configuration and cross-configuration settings. Our results show that while high classification accuracy can be achieved under controlled conditions, performance degrades significantly when models are evaluated across different network configurations, particularly for fine-grained identification tasks. Finally, we investigate the trade-off between traffic storage requirements and classification accuracy. We show that lossy compression of traffic features through quantization can reduce storage requirements by approximately 4–5× compared to lossless storage of raw packet traces, while preserving near-lossless classification performance. Overall, our results highlight the need for topology-aware Zigbee traffic analysis and storage-efficient feature compression to enable robust and scalable IoT forensic systems.

Keywords: Internet of Things, IoT Forensics, Zigbee, Encrypted Network Traffic Analysis, Network Traffic Compression

1. Introduction

The Internet of Things (IoT) has become an integral component of contemporary society, with Consumer IoT (CIoT) devices accounting for nearly 50% of the overall IoT market [1]. Although these technologies provide substantial benefits in everyday contexts, their pervasive deployment across domestic, professional, and public environments—together with their persistent connection to network infrastructures—raises significant concerns related to privacy and security. In this context, smart home devices are of particular interest, as they continuously and often unobtrusively collect data related to users' daily activities. Such characteristics make them a valuable source for IoT forensics, enabling the acquisition and analysis of device-generated data, including network traffic, which can be leveraged to infer events, behaviors, or usage patterns [2]. Two communication protocols are predominantly used for interconnecting smart home devices: Wi-Fi and Zigbee. While a substantial body of literature has demonstrated the feasibility of IoT forensic tools for capturing and analyzing traffic generated by Wi-Fi–enabled devices [3, 4, 5], forensic methodologies specifically targeting Zigbee-based devices remain comparatively underexplored. This gap is largely due to the inherent challenges associated with Zigbee traffic analysis. First, the characteristics of the Zigbee protocol, which departs significantly from the conventional TCP/IP stack, introduce additional complexity in the interpretation of captured traffic. Furthermore, Zigbee networks typically generate smaller packet sizes and lower traffic volumes than Wi-Fi, limiting the amount of observable information available for forensic inference. Finally, Zigbee commonly relies on a mesh network topology, in which devices forward traffic on behalf of others rather than communicating directly with a central access point, as is the case with Wi-Fi. This architectural feature further complicates traffic analysis, as relay operations introduce noise into traffic traces and such noise is inherently time-varying due to dynamic routing path adaptations. As a result, forensic insights derived from traffic analysis in a given Zigbee network topology may not generalize to other topologies, in contrast to what is more commonly observed in Wi-Fi–based environments. This study addresses the aforementioned challenges through the following main contributions:

- *ZIOTP2026 traffic dataset*[1]*:* We collected and publicly released a comprehensive dataset of Zigbee network traffic captured from 21 off-the-shelf smart home devices deployed under two distinct network configurations. The dataset includes both background traffic and traffic generated through controlled user

[1] https://github.com/antonio-boiano/ZIOTP2026

interactions, and is fully labeled to support IoT forensic tasks such as device identification, event detection, and traffic classification. To the best of the authors' knowledge, this represents the largest publicly available labeled Zigbee traffic dataset specifically designed for IoT forensics.

- *Traffic analysis:* Leveraging a State of the Art (SoA)-based methodology, we developed a Machine Learning (ML) framework for Zigbee device classification, addressing two core tasks: device type classification and individual device identification. To mitigate the complexity introduced by multi-hop communications in mesh network topologies—where devices forward packets on behalf of others—we restrict the analysis to first-hop messages, thereby reducing bias due to retransmission and relaying behavior. Nevertheless, we show that temporal traffic features remain strongly affected by topology-dependent dynamics. Experimental results demonstrate that, while intra-topology classification achieves excellent performance (with F1-scores above 0.95), cross-topology generalization suffers from significant degradation. This highlights the critical and often overlooked impact of Zigbee mesh topology on the generalizability of traffic-based classification models.

- *Storage–accuracy analysis:* We investigate the trade-off between traffic storage requirements and the accuracy of downstream forensic analysis tasks. Specifically, we compare lossless traffic compression—obtained by applying state-of-the-art compression algorithms directly to raw Zigbee traffic—with lossy compression of traffic representations, where scalar quantization is applied to the statistical features used as input to ML models. Our results show that the latter approach enables a substantial reduction in storage bitrate without degrading classification accuracy, making it a promising solution for IoT forensic systems tasked with long-term storage of large-scale traffic traces.

The remainder of this paper is organized as follows. Section 2 introduces the Zigbee protocol architecture, while Section 3 reviews related work in IoT forensics and Zigbee traffic analysis. Section 4 presents the ZIOTP2026 dataset and the adopted data collection methodology. Section 5 describes the feature extraction and classification framework, and Section 6 discusses the experimental results under both intra-topology and cross-topology settings. Section 7 analyzes the trade-off between traffic storage and classification accuracy. Finally, Section 8 concludes the paper and outlines directions for future research.

2. The Zigbee Protocol

Zigbee is a low-cost, low-power wireless communication protocol designed for short-range communications, making it well suited for embedded systems and smart home applications. Its support for mesh networking enables the deployment of robust and self-healing networks with extended coverage. Zigbee is built on top of the IEEE 802.15.4 standard [6], which specifies low-rate wireless personal area networks (LR-WPANs) optimized for low data rates and minimal energy consumption.

2.1. Zigbee Network Architecture

The Zigbee protocol stack consists of four layers: the physical (PHY) layer, the medium access control (MAC) layer, the network (NWK) layer, and the application (APL) layer. The PHY and MAC layers are defined by IEEE 802.15.4 and are responsible for physical transmission and channel access control.

IEEE 802.15.4 operates over multiple frequency bands to accommodate regional regulations. In the widely adopted 2.4 GHz band, it supports 16 channels with 5 MHz spacing. The MAC layer employs a contention-based access mechanism based on CSMA/CA. In beacon-enabled networks, the PAN Coordinator periodically transmits beacon frames containing network parameters and synchronization information.

Building on IEEE 802.15.4, Zigbee introduces the NWK and APL layers to enable multi-hop routing, network management, and application-level services. The application layer includes the Application Support Sublayer (APS), the Zigbee Device Object (ZDO), and an application framework defining device-specific operations. The APS handles data binding, fragmentation, and security, while the ZDO is responsible for network formation, device discovery, and security configuration. Although Zigbee application profiles promote interoperability, their standardized data formats can increase the complexity of traffic-based device classification.

Zigbee networks comprise three main device roles: the PAN Coordinator (PANC), Full Function Devices (FFDs), and Reduced Function Devices (RFDs). The PANC manages network creation, addressing, and coordination. FFDs, commonly acting as routers, relay packets on behalf of other devices and enable multi-hop communication, which allows Zigbee networks to extend coverage without relying on a high-power central access point. RFDs are typically low-power sensors or actuators that communicate through routers or directly with the PANC.

Each Zigbee device is uniquely identified by a 64-bit IEEE MAC address (extended address). After joining the network, a 16-bit short address is assigned by the coordinator and used for subsequent intra-network communications.

These architectural characteristics have direct implications for traffic observability and analysis, particularly in forensic scenarios involving multi-hop communications and dynamically evolving network topologies.

3. State of the Art

This section reviews prior work related to the main contributions of this study, covering the availability of Zigbee network traffic datasets, existing approaches to Zigbee traffic analysis and profiling, and prior efforts on the trade-off between traffic storage requirements and analysis accuracy, which have so far been explored primarily in Wi-Fi and IP-based network settings.

3.1. Dataset Availability

Recent survey works provide a comprehensive overview of publicly available network traffic datasets for IoT systems, consistently highlighting the severe scarcity of datasets containing realistic Zigbee traffic. In particular, out of 74 datasets surveyed in [7], only seven include Zigbee traffic, and only a small subset of these provide packet-level captures suitable for fine-grained traffic analysis, such as device fingerprinting or physical-layer studies.

Among the publicly available datasets, the work in [8] is the only one that includes traffic from ten commercial Zigbee devices commonly deployed in smart home environments. Traffic was captured using a software-defined radio platform (USRP N210) with GNU Radio and the gr-ieee802-15-4 and gr-foo modules, enabling IEEE 802.15.4 packet reception and storage in PCAP format. However, the dataset is released exclusively as unlabeled raw traffic traces, limiting its applicability to supervised IoT forensic tasks.

Most other studies rely on proprietary or non-public datasets. For instance, Sadikin et al. [9] developed a Zigbee intrusion detection system and collected attack traces involving one gateway and three smart light bulbs, but did not release the dataset. Bihl et al. [10] conducted physical-layer fingerprinting experiments on ten identical Texas Instruments CC2420 Zigbee devices by collecting RF emissions in controlled environments. Patel et al. [11] extended this line of work to 13 CC2420 devices and evaluated ensemble learning techniques, such as Random Forest and AdaBoost, achieving improved identification performance compared to parametric classifiers. Despite their promising results, the datasets used in these studies are not publicly available.

Among the few public datasets that include Zigbee traffic, Dadkhah et al. [12] introduced a multidimensional IoT profiling dataset primarily focused on IP-based

devices, with a limited Zigbee subset. This subset includes nine distinct Zigbee devices (16 instances) from four manufacturers, captured under controlled laboratory conditions. While noteworthy for incorporating non-IP IoT technologies, the Zigbee component of the dataset exhibits limited interaction diversity, as only single, non-repetitive device–action pairs were recorded. Such constraints limit its suitability for comprehensive traffic profiling in complex and distributed environments such as smart homes. Lourme et al. [13] and Keleşoğlu et al. [14] released ZBDS2023 and ZigBeeNet, respectively: multi-day Zigbee traffic traces collected from 10 and 14 devices, with RSSI information available. ZBDS2023 further provides multi-point captures using multiple sniffers deployed at different locations within the home. Despite these strengths, both datasets offer limited support for evaluating generalization in realistic smart-home settings, as the devices are largely drawn from a single vendor ecosystem (predominantly Philips) and the captures do not span multiple network topologies, preventing robustness assessments of ML models across diverse deployment conditions.

Motivated by these limitations, the present work introduces a dedicated Zigbee data acquisition campaign aimed at increasing both interaction diversity and traffic volume. In contrast to prior datasets, our campaign captures traffic from 21 Zigbee devices spanning multiple functional classes, including temperature sensors, door sensors, motion detectors, smart plugs, smart bulbs, vibration sensors, and light sensors, and collects data in multi-topology environments where devices operate concurrently rather than in isolation. This setup enables a systematic analysis of how network topology and deployment conditions affect the performance and generalization of state-of-the-art traffic-profiling methodologies.

3.2. Zigbee Network Traffic Analysis

Despite the extensive literature on IoT traffic classification, relatively few studies focus specifically on Zigbee-based traffic analysis. Rong Li et al. [15] investigated potential privacy leakage from encrypted Zigbee traffic, aiming to infer device type and operational state. Their approach extends classical feature-based traffic analysis by incorporating inter-event timing information, along with selected protocol-level features such as MAC addresses and channel identifiers. Machine learning models were then employed for device classification, identification, and event detection.

Similarly, the authors of [16] proposed a fingerprinting framework for the passive identification of device classes in Zigbee and Z-Wave networks. Their method primarily relies on inter-arrival time (IAT) statistics of network frames, achieving effective device-class identification. Other works, such as [17, 18], also explored Zigbee traffic analysis, but were limited by the use of small-scale datasets.

Additional studies have examined encrypted Zigbee traffic from different perspectives. Some focus on exploiting protocol vulnerabilities to infer sensitive device information, such as logical device roles [19] or spatial location [20]. Others have released Zigbee datasets targeting either device profiling [12] or security applications, such as the dataset introduced in [21] for intrusion detection. In parallel, several works have addressed Zigbee device fingerprinting using physical-layer characteristics rather than network traffic features [22, 23, 24].

While existing studies demonstrate the feasibility of Zigbee traffic classification, they generally overlook the impact of mesh network topology on model robustness. Unlike IP-based networks, Zigbee extensively relies on multi-hop communication, where changes in network topology may significantly affect observed traffic patterns. In this work, we build upon methodologies proposed in the state of the art to systematically quantify performance degradation across different network topologies, providing a baseline for machine-learning-based Zigbee traffic profiling in realistic large-scale smart home environments.

3.3. Storage-accuracy analysis

The trade-off between traffic storage requirements and the accuracy of downstream forensic analysis has received limited attention in IoT forensics, where most approaches assume the availability of complete traffic traces and overlook long-term storage scalability issues. An initial study of this problem was presented in [4, 5], which proposed a framework for optimizing storage efficiency while preserving analysis accuracy. Focusing on Wi-Fi and IP-based IoT traffic, the authors compared lossless compression of raw traces with lossy compression of extracted traffic features, showing that feature-level compression can substantially reduce storage requirements without degrading classification performance. More recently, Palmese et al. [25] investigated resource optimization for evidence collection and preservation in forensics-ready access points, analyzing how storage and processing constraints affect IP-based traffic monitoring at the network edge. Despite these contributions, existing works remain limited to Wi-Fi and IP-based technologies and do not consider the distinctive characteristics of Zigbee networks, such as low traffic rates, small packet sizes, and mesh-based multi-hop communication. This work extends the storage–accuracy analysis to Zigbee traffic, evaluating compression strategies in realistic multi-topology smart home scenarios.

4. ZIOTP2026: A Dataset for Zigbee traffic classification

Despite the widespread adoption of Zigbee in smart home environments, publicly available Zigbee traffic datasets remain scarce. To address this gap and support re-

producible research, we introduce ZIOTP2026, a dataset of Zigbee network traffic collected over multiple days in realistic smart home settings and capturing natural user-device interactions. The data collection methodology builds upon prior work [12], extending its Zigbee component by capturing traffic from 21 off-the-shelf devices spanning 15 models and seven manufacturers. To enable the investigation of topology-dependent effects, devices were deployed across two distinct Zigbee personal area networks (PANs) during data acquisition. This multi-topology design allows systematic analysis of how network configuration impacts traffic-based forensic tasks such as device classification and identification.

4.1. Acquisition Setup

A dedicated network architecture was deployed to replicate a realistic smart home environment. An isolated Local Area Network (LAN) was configured using a consumer-grade router running OpenWRT, with a Raspberry Pi running Home Assistant OS [26] acting as the central smart home gateway and Zigbee PANC. Zigbee connectivity was provided through a USB radio interface managed by the Zigbee Home Automation (ZHA) integration[2], which handled network formation and device onboarding. For all experiments, the Personal Area Network (PAN) was configured on Channel 11 and devices were added in isolation to ensure reproducible initialization.

IP-based traffic exchanged within the LAN was captured from a host connected to the network, which also provided access to the Home Assistant dashboard. Zigbee traffic at the IEEE 802.15.4 layer was collected using a dedicated packet sniffer operating in promiscuous mode. To ensure reliable capture, the sniffer and the Zigbee radio interface were placed in close proximity. The overall network configuration is illustrated in Figure 1.

To analyze the impact of network topology on Zigbee traffic analysis, data acquisition was conducted under two distinct Zigbee network topologies. Here, topology refers to the spatial arrangement of devices and the resulting communication paths among the PANC, Full Function Devices (FFDs), and Reduced Function Devices (RFDs). Unlike Wi-Fi's star topology, Zigbee relies on mesh communication, where messages may be forwarded through intermediate FFDs. The two adopted topologies, shown in Figure 2, differ in device placement and coordinator position, inducing distinct multi-hop communication patterns. The acquisition procedure was kept identical across both topologies.

[2]https://www.home-assistant.io/integrations/zha/

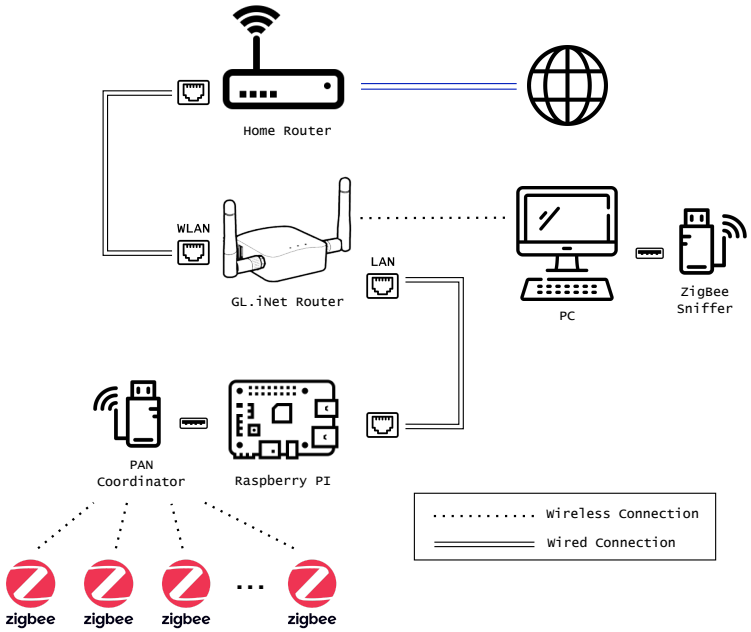


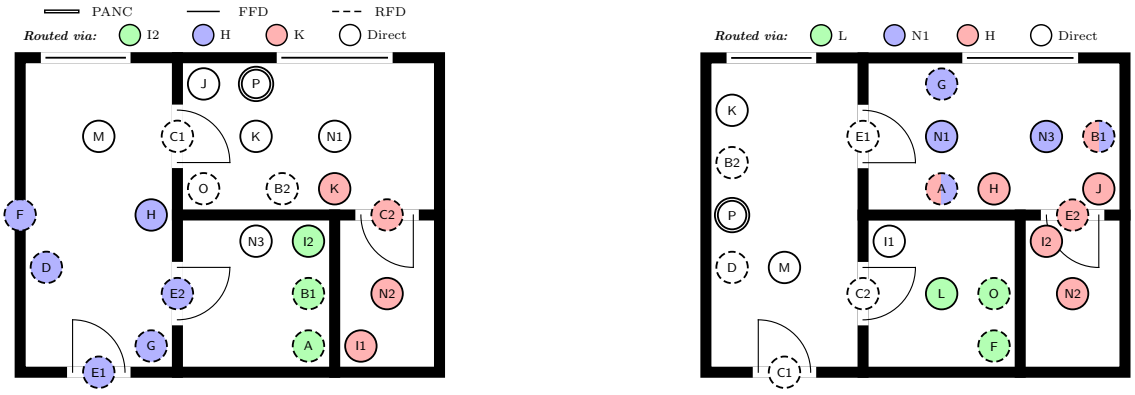
Figure 1: Network configuration of the smart home and acquisition pipeline used to capture Zigbee and IP traffic from commercial IoT devices. The smart home is managed by a central gateway acting as PAN coordinator, while a dedicated sniffer and a monitoring host capture Zigbee and IP network traces within an isolated network.

4.2. Acquiring Device Behaviors

Following the acquisition methodology introduced in [12], a set of controlled interaction scenarios was designed to capture a wide range of Zigbee device behaviors and operational states. The acquisition campaign included: (i) idle periods with no user interaction, used to characterize background and periodic traffic; (ii) power cycling events for mains-powered devices, capturing network join and leave procedures; (iii) user-issued commands executed through the smart home controller; (iv) direct physical interactions with sensors (e.g., motion, door, vibration); and (v) extended active scenarios emulating natural day-to-day smart home usage. Each scenario was repeated multiple times to capture temporal variability and enrich the dataset with diverse traffic patterns.

4.3. Ground Truth Extraction

Since the dataset captures encrypted Zigbee traffic, device identity and application-layer payloads are not directly observable. Device identities were recovered a posteriori by isolating each device and associating its traffic with the corresponding IEEE



(a) Topology A

(b) Topology B

Figure 2: Device disposition during the acquisition campaign for the two considered topologies. Devices are positioned according to their physical deployment and labeled as reported in Table 1. The outer line style identifies the device role (PANC, FFD, or RFD). Color coding indicates the logical communication path toward the PANC: white denotes devices directly associated with the PANC, while colored devices communicate with the PANC via an intermediate FFD acting as a router.

MAC address. To enable event-level labeling, each device was individually reset and paired with the PANC, while Zigbee traffic exchanged during the pairing phase was captured.

During commissioning, the transport key is exchanged within the Zigbee Application Support (APS) layer and encrypted using a publicly known trust center link key. Leveraging this mechanism, pairing traffic was decrypted using Wireshark[3] to extract the device-specific transport key, which was then used exclusively for dataset labeling and validation.

All subsequent traffic analysis and machine learning experiments were performed on encrypted traffic only, without relying on decrypted payload information.

4.4. Dataset Characteristics and Availability

The ZIOTP2026 dataset comprises 252,896 IEEE 802.15.4 frames, including 69,497 Zigbee frames corresponding to approximately 2,520 human-generated commands collected across two network topologies. The traffic volume is nearly balanced between the two configurations, with Topology B containing approximately 3% more data than Topology A. Each topology-specific dataset occupies approximately 17 MB and includes traffic generated by the PANC, mains-powered devices (e.g., smart

[3]https://www.wireshark.org/

Label	Brand	Device	Category	Model	Qty	Type	Data
A	Sonoff	Temperature Sensor	Temperature	SNZB-02	1	RFD	0.
B	Sonoff	Motion Sensor	Motion	SNZB-03	2	RFD	0.
C	Sonoff	Door Sensor	Door	SNZB-04	2	RFD	0.
D	Aqara	Motion Sensor P1	Motion	MS-S02	1	RFD	0.
E	Aqara	Door and Window Sensor	Door	DW-S02D	2	RFD	0.
F	Aqara	Vibration Sensor	Motion	DJT11LM	1	RFD	0.
G	Aqara	Mini Switch	Button	WB-R02D	1	RFD	0.
H	Tuya	Smart Socket EU-16A	Socket	N/A	1	FFD	16.
I	Senckit	Smart Power Plug	Socket	N/A	2	FFD	15.
J	Ledvance	Z3 Plug	Socket	4058075208513	1	FFD	3.
K	Ledvance	Smart+ Plug	Socket	4058075729261	1	FFD	2.
L	Ledvance	Smart+ Bulb	Bulb	4058075208391	1	FFD	8.
M	Moes	Bulb	Bulb	GU10	1	FFD	4.
N	Philips	HueLamp	Bulb	LWA019	3	FFD	5.
O	Philips	HueMotion	Motion	SML001	1	RFD	4.

Table 1: List of Zigbee IoT devices included in the dataset, with information on: brand, device type, model identifier, and quantity. Device identifiers are assigned systematically using alphabetical labels, with numerical suffixes appended when multiple units of the same device type are present. The table additionally presents the per-device data distribution, quantifying each device's proportional contribution to the total volume of Zigbee packets captured in the two topologies.

bulbs and plugs), and battery-operated devices. In addition, around 40 MB of IP-based traffic captured during command interaction phases is provided, enabling cross-technology analysis. The dataset includes traffic from 21 commercial Zigbee devices spanning multiple categories (e.g., sensors, actuators, and lighting devices) and several manufacturers. Multiple instances of selected devices are included to support comparative analysis across devices of the same model. Table 1 summarizes device models, quantities, and per-device traffic contributions. As expected, traffic volume is highly unbalanced across device roles, with RFDs generating up to two orders of magnitude fewer packets than FFDs. This imbalance reflects the low-duty-cycle design of battery-powered devices and introduces additional challenges for machine-learning-based traffic analysis. The ZIOTP2026 dataset is publicly available to support reproducible research in Zigbee traffic analysis and IoT forensics.

5. Traffic Analysis Methodology

This section describes the methodology adopted to analyze encrypted Zigbee traffic and evaluate traffic-based device profiling under different network topologies. Two classification tasks are considered: *device type classification* and *ndividual device identification*. The former targets coarse-grained categorization of devices based on functional roles and communication patterns, while the latter addresses fine-grained discrimination among individual devices. Although Zigbee enforces standardized message formats, device-specific characteristics remain observable due to differences in implementation, hardware, and operational behavior.

5.1. Data Preprocessing

Raw Zigbee traffic captured in PCAP format was processed through a preprocessing pipeline designed to generate structured representations suitable for machine-learning-based analysis.

Ground truth and metadata extraction. Each packet was associated with device-level metadata by mapping 16-bit Zigbee short addresses to device categories and labels (Table 1). Payload decryption was used exclusively during dataset labeling to distinguish human-triggered interactions from background traffic, while all subsequent analysis relied solely on encrypted traffic. Packet-level attributes extracted include source and destination identifiers at the IEEE 802.15.4 and Zigbee NWK layers, packet length, and timestamp.

Filtering and feature extraction. To capture temporal dependencies, packet traces were segmented into non-overlapping time windows, each aggregating consecutive packets. For each device, bidirectional traffic exchanged with the PANC was isolated, and relayed packets introduced by mesh routing were filtered by comparing Zigbee-layer source addresses with IEEE 802.15.4 transmitter addresses.

For each window containing at least two packets, statistical features were extracted from packet lengths and inter-arrival times, computed separately for uplink, downlink, and aggregated traffic. Extracted metrics include mean, standard deviation, minimum, maximum, median, skewness, kurtosis, mean absolute deviation, and packet count.

5.2. Classification and Validation Strategy

Both classification tasks were addressed using eXtreme Gradient Boosting (XGBoost), which is well suited for multi-class classification on structured feature sets. Hyperparameters were optimized independently for each task using grid search with cross-validation.

For device type classification, the model employed 300 estimators with a maximum tree depth of 8 and moderate regularization. Individual device identification required a more expressive configuration, using the same number of estimators but deeper trees (maximum depth of 10) and additional instance subsampling to improve robustness. All remaining parameters were left at default values.

Model evaluation was performed using stratified 5-fold cross-validation to preserve class distributions. Performance was assessed using both weighted and macro-averaged F1-scores. Unless otherwise stated, results are reported using the macro F1-score, which is particularly sensitive to performance on minority classes and therefore suitable for the imbalanced nature of the dataset.

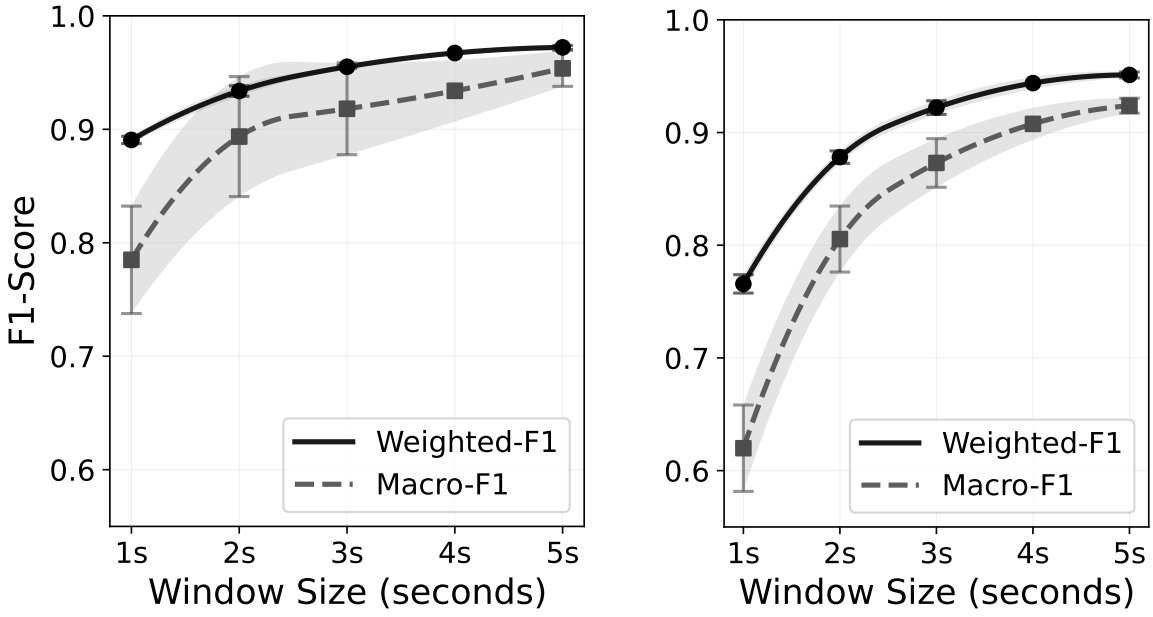
6. Performance Evaluation

This section evaluates the performance of the traffic analysis methodology introduced in Section 5. We assess the effectiveness of the XGBoost-based classifier for the two traffic analysis tasks considered in this work—device type classification and individual device identification—under both intra-topology and cross-topology evaluation settings.

The evaluation is structured as follows. First, we perform a preliminary analysis to identify the optimal temporal window size for feature extraction. We then report classification performance when training and testing are conducted within the same network topology, representing the standard evaluation scenario adopted in the literature. Finally, we analyze the generalization capability of the model by evaluating performance across different network topologies.

6.1. Optimal Time Window Selection

Before proceeding with the main classification experiments, we conducted a preliminary study to determine the optimal temporal window size for feature extraction. Since our approach aggregates network traffic data using time windows, this parameter critically affects the trade-off between capturing sufficient temporal context and maintaining adequate sample sizes for robust classification. We tested five different window lengths (1, 2, 3, 4, and 5 seconds), training and evaluating the XGBoost classifier for each configuration. Figure 3 presents the weighted and macro F1-scores across these window sizes, revealing a consistent trend for both tasks: classification performance improves steadily with increasing window size, as longer windows more effectively capture meaningful temporal patterns in network traffic data. Specifically, for device type classification, the weighted F1-score increased from 0.89 (1-second window) to 0.97 (5-second window), and the macro F1-score improved from 0.78



(a) Device Type Classification

(b) Individual Device Identification

Figure 3: Weighted and Macro F1-Score across different window sizes for: (a) device type classification and (b) individual device identification classification task.

to 0.95. For individual device identification, in the same range of window sizes, the weighted F1-score increased from 0.76 to 0.95, and the macro F1-score from 0.62 to 0.92. Both tasks exhibit performance curves approximating a saturating exponential function, characterized by rapid initial gains followed by progressively diminishing returns. The marginal improvement beyond the 4-second threshold is minimal, suggesting that a 5-second window captures nearly all relevant temporal dynamics. Based on this empirical characterization and considering the trade-off between classification performance, computational efficiency, and prediction time, we fixed the sliding window size at 5 seconds for all subsequent experiments presented in the next sections.

Before presenting the classification results, a statistical analysis of the ML-ready dataset (described in Section 5.1) revealed that RFD devices, which typically transmit single Zigbee packets in response to specific events, can produce windows containing insufficient packets (i.e., windows with fewer than 2 packets) for meaningful feature extraction. Consequently, two devices: the Sonoff temperature sensor and the Aquara Vibration sensor, yielded fewer than 10 valid 5-second windows across both topologies and were excluded from further analysis.

Based on these results, a 5-second temporal window was selected for all subse-

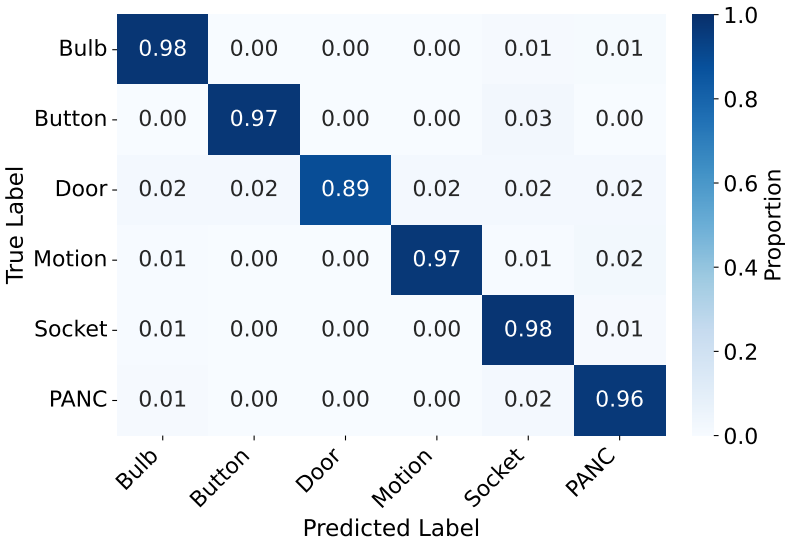


Figure 4: Normalized Confusion matrix for device type classification on a 5-second window in topology B.

quent experiments.

6.2. Intra-topology Classification

We first evaluate classification performance under an intra-topology setting, where training and testing data originate from the same Zigbee network topology. This scenario represents the standard evaluation setup adopted in prior work on IoT traffic classification and serves as a baseline for subsequent cross-topology analysis. All results are obtained using stratified 5-fold cross-validation.

6.2.1. Device Type Classification

The proposed classifier achieves excellent performance when evaluated under the intra-topology setting. Using a 5-second temporal window and topology B, the model attains a weighted F1-score of 0.97 and a macro F1-score of 0.95. The corresponding classification report and normalized confusion matrix are shown in Table 2 and Figure 4, respectively.

High precision and recall are consistently observed across nearly all device categories, indicating strong discriminative capability between different smart home device types. Comparable results are obtained for topology A, as summarized in Table 4.

Class	Precision	Recall	F1-score
Bulb	0.94	0.98	0.96
Button	0.92	0.97	0.95
Coordinator	0.98	0.96	0.97
Door	0.98	0.89	0.93
Motion	0.92	0.97	0.94
Socket	0.98	0.98	0.98
Accuracy			0.97
Macro Avg	0.95	0.96	0.95
Weighted Avg	0.97	0.97	0.97

Table 2: Classification report for device type classification on a 5-second window in topology B.

6.2.2. Individual Device Identification

Despite the increased complexity of individual device identification, the classifier maintains strong performance under the intra-topology setting. With 14 distinct device identities, the model achieves a weighted F1-score of 0.95 and a macro F1-score of 0.93 when evaluated on topology B.

The normalized confusion matrix and classification report, shown in Figure 5 and Table 3, respectively, indicate robust performance across most devices. The PANC is particularly well distinguished, likely due to its unique role in network coordination. Results for both topology A and B are reported in Table 5.

6.3. Cross-topology Classification

We next evaluate the generalization capability of the proposed traffic analysis methodology by adopting a cross-topology setting, in which training and testing are performed on data collected from different Zigbee network topologies. This evaluation exploits the multi-topology nature of the dataset and allows us to quantify the impact of topology-dependent traffic dynamics on classification performance.

6.3.1. Device Type Classification

Device type classification exhibits moderate performance degradation under cross-topology evaluation. When trained on one topology and tested on the other, the model achieves a weighted F1-score of approximately 0.85 and a macro F1-score of 0.75 (Table 4).

While the PANC remains consistently well classified, device categories dominated by RFDs experience larger performance drops. This behavior is attributable to

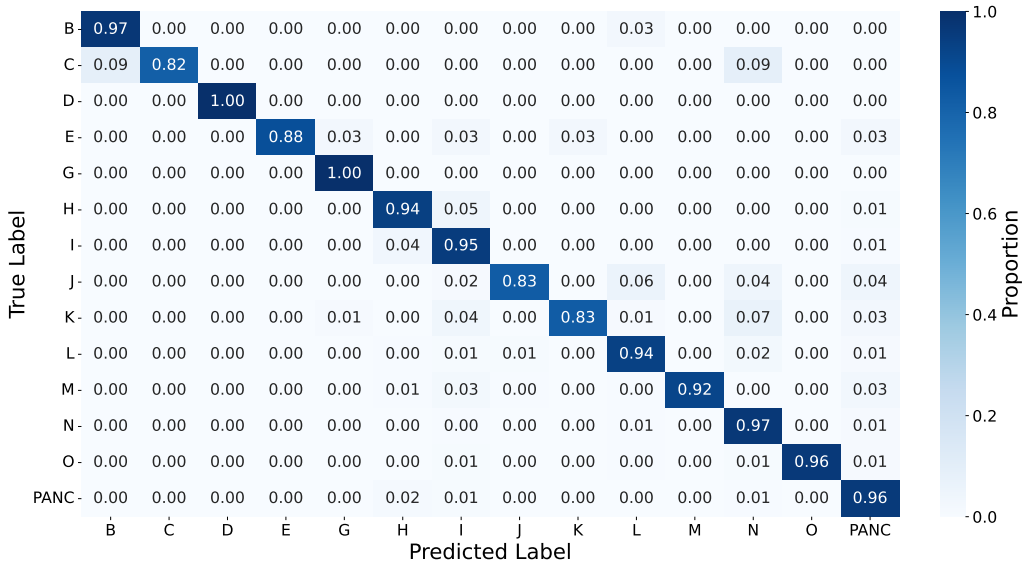


Figure 5: Normalized Confusion matrix for individual device identification on a 5-second window in topology B.

limited training support and to topology-dependent routing effects introduced by the Zigbee mesh network. Overall, the classifier remains effective at distinguishing device types, reflecting the standardized command sets and communication patterns enforced by the Zigbee protocol.

6.3.2. Individual Device Identification

In contrast, individual device identification is strongly affected by topology changes. When trained on topology B and tested on topology A, accuracy drops to 0.68, with a weighted F1-score of 0.67 and a macro F1-score of 0.41 (Table 5).

The most significant degradation is observed for RFDs, whose average per-device F1-score falls below 0.2. This effect is exacerbated by the fine-grained nature of the task, which increases the number of classes and reduces the number of training samples per class. FFDs demonstrate comparatively higher robustness, with an average F1-score of 0.53, while the PANC maintains stable performance (F1-score of 0.86), indicating largely topology-invariant traffic patterns.

Misclassifications predominantly occur among devices of the same manufacturer or device category, suggesting that manufacturer-specific traffic characteristics exhibit partial resilience to topology changes, whereas device-specific fingerprints are more sensitive to mesh routing dynamics.

Class	Class Name	Precision	Recall	F1-score
B	Sonoff Motion Sensor	0.84	0.97	0.90
C	Sonoff Door Sensor	0.90	0.82	0.86
D	Aqara Motion Sensor	1.00	1.00	1.00
E	Aqara Door/Window Sensor	1.00	0.88	0.94
G	Aqara Mini Switch	0.92	1.00	0.96
H	Tuya Smart Socket	0.91	0.94	0.92
I	Senckit Smart Power Plug	0.95	0.95	0.95
J	Ledvance Z3 Plug	0.87	0.83	0.85
K	Ledvance Smart+ Plug	0.88	0.83	0.85
L	Ledvance Smart+ Bulb	0.94	0.94	0.94
M	Moes Bulb	0.95	0.92	0.94
N	Philips HueLamp	0.93	0.97	0.95
O	Philips HueMotion	0.92	0.96	0.94
PANC	Pan Coordinator	0.98	0.96	0.97
Accuracy				0.95
Macro Avg		0.93	0.93	0.93
Weighted Avg		0.95	0.95	0.95

Table 3: Classification report for individual device identification using a 5-second temporal window in topology B.

6.4. Discussion

The experimental results highlight the effectiveness and the limitations of traffic-based device profiling in Zigbee networks. Under intra-topology conditions, the proposed methodology achieves excellent performance for both device type classification and individual device identification, confirming that encrypted Zigbee traffic retains sufficient behavioral information to support fine-grained analysis.

However, the cross-topology evaluation reveals a markedly different behavior between the two tasks. While device type classification shows only moderate performance degradation, individual device identification is significantly affected by changes in network topology. This divergence can be attributed to the standardized nature of Zigbee application profiles, which enforces consistent communication patterns at the device category level, in contrast to device-specific characteristics that are more sensitive to mesh routing dynamics and multi-hop communication paths.

The observed degradation is particularly pronounced for battery-powered RFDs, whose sparse and event-driven traffic, combined with relay operations performed by

Table 4: Classification Results Summary for Device Type classification task

Train Dataset	Test Dataset	Accuracy	Precision	Recall	F1-Macro	F1-Weighted
Intra-Topology (K-Fold, k=5)						
Topo. A	Topo. A	0.970 ± 0.003	0.949 ± 0.013	0.944 ± 0.022	0.945 ± 0.010	0.971 ± 0.003
Topo. B	Topo. B	0.972 ± 0.002	0.953 ± 0.018	0.957 ± 0.013	0.954 ± 0.016	0.972 ± 0.002
Cross-Topology (Full Dataset)						
Topo. A	Topo. B	0.878	0.738	0.765	0.745	0.879
Topo. B	Topo. A	0.850	0.724	0.759	0.741	0.849

Table 5: Classification Results Summary for individual device identification task

Train Dataset	Test Dataset	Accuracy	Precision	Recall	F1-Macro	F1-Weighted
Topo. A	Topo. A	0.944 ± 0.004	0.910 ± 0.032	0.877 ± 0.017	0.889 ± 0.021	0.944 ± 0.004
Topo. B	Topo. B	0.951 ± 0.003	0.932 ± 0.016	0.924 ± 0.019	0.923 ± 0.007	0.951 ± 0.003
Cross-Topology (Full Dataset)						
Topo. A	Topo. B	0.692	0.517	0.480	0.474	0.694
Topo. B	Topo. A	0.678	0.446	0.434	0.413	0.672

FFDs, reduces the stability of temporal traffic features across topologies. Conversely, mains-powered devices and the PANC exhibit more robust and topology-invariant traffic patterns, resulting in higher classification resilience.

7. Storage–Accuracy Analysis

Captured Zigbee network traffic must often be retained to support a-posteriori IoT forensic investigations and long-term monitoring. However, the limited storage resources typically available on edge IoT controllers, combined with the continuous operation of smart home devices, make efficient traffic storage a critical design requirement.

In this section, we analyze the trade-off between storage efficiency and classification accuracy for Zigbee traffic analysis. We compare lossless compression techniques applied directly to raw packet traces with lossy approaches based on compact representations of traffic features. The goal is to quantify how different storage strategies impact the accuracy of downstream traffic classification tasks, and to identify solu-

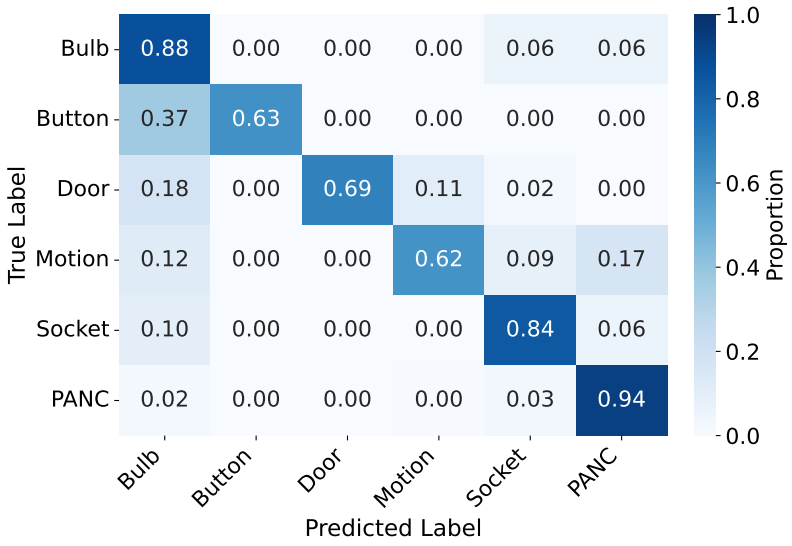


Figure 6: Normalized Confusion matrix for device type classification on cross-topology settings (trained on network traces acquired from topology B and tested against network traces coming from topology A.

tions that enable substantial storage reduction without degrading forensic analysis performance.

7.1. Lossless Compression of Raw Zigbee Packets

We first evaluate the feasibility of long-term storage of captured Zigbee traffic by retaining raw packet traces in `.pcap` format and applying standard lossless compression techniques. This approach represents the most straightforward solution for traffic preservation, as it preserves full packet-level information and does not require any modification to the original data representation. At the same time, it establishes an upper bound on storage efficiency achievable without sacrificing information content. Table 6 reports the storage footprint and average bitrate required to store raw Zigbee traffic captured under the two considered network topologies, together with the compression performance achieved using three widely adopted lossless compressors: GZIP, BZIP2, and 7Z-LZMA. These tools were selected due to their protocol-agnostic nature and widespread use in network trace archival.

Without compression, raw Zigbee traffic requires average storage rates of approximately 2.2 kbps and 2.9 kbps for Topology A and Topology B, respectively. Here, the storage rate is computed as the total size of the stored `.pcap` file divided by the corresponding capture duration, yielding an average bitrate over time. These

values reflect the cumulative effect of long capture durations and continuous background traffic, despite the low data rate and small packet size characteristic of Zigbee communications.

Applying lossless compression substantially reduces storage requirements. GZIP and BZIP2 achieve comparable performance, reducing the bitrate by approximately 64–65% across both topologies. Among the evaluated methods, 7Z-LZMA consistently provides the best compression performance, lowering the required storage rate to 657 bps for Topology A and 768 bps for Topology B, corresponding to an approximate 3.3× compression ratio relative to uncompressed `.pcap` files.

While lossless compression substantially reduces the storage footprint of raw Zigbee traffic, the resulting storage rates can still pose practical constraints in edge-based deployments characterized by limited storage resources and long-term evidence retention requirements. This motivates the exploration of alternative storage strategies based on compact traffic representations and controlled information loss.

Table 6: Evaluation of Lossless Compression Methods on the ZIOTP2026 PCAP Dataset: Size and Bitrate Across Two Network Topologies

	Topology A		Topology B	
Method	Size (MB)	bps	Size (MB)	bps
Raw	6.69	2180	11.06	2880
GZIP	2.41	786	3.60	937
BZIP2	2.36	771	3.51	913
7Z-LZMA	2.02	657	2.95	768

7.2. Storage of Traffic Features

An alternative to storing raw packet traces consists in retaining compact representations of traffic behavior in the form of extracted features. This approach is particularly attractive when the primary objective is traffic analysis rather than packet-level reconstruction, as it eliminates the need for repeated feature extraction during offline processing.

However, unlike IP-based networks, Zigbee environments are characterized by low traffic intensity and small packet sizes. As a consequence, storing raw Zigbee packet traces—especially when combined with lossless compression—may already be relatively storage-efficient. In this setting, feature-based representations are not intrinsically more compact and may require higher storage rates when features are stored using standard high-precision formats.

To clarify this aspect, we consider the window-based feature extraction process described in Section 5.1. Let N denote the number of devices being monitored, F the number of extracted features per window, b the bit width used to encode each feature, and T the window duration. Under these assumptions, the resulting average storage rate can be expressed as

$$\frac{N \times F \times b}{T} \quad \text{bit/s}. \tag{1}$$

This relationship highlights that, in Zigbee-based systems, the storage cost of feature-level representations can easily exceed that of raw packet traces when high-precision feature encodings are adopted.

These considerations indicate that feature-level storage does not automatically yield storage savings in Zigbee networks. Instead, any potential benefit must arise from reducing the precision of the feature representation. To this end, we investigate lossy compression of traffic features through scalar quantization, where each feature is encoded using a reduced number of bits.

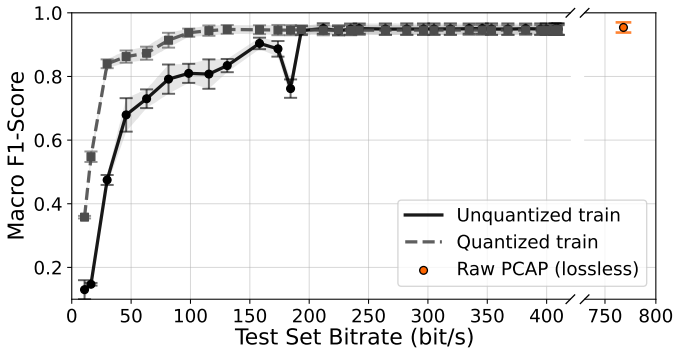
Specifically, we generate quantized feature representations using bit depths ranging from 1 to 64 bits per feature. The resulting quantized feature vectors are subsequently compressed using a lossless encoder (7Z-LZMA) to remove residual redundancy. We evaluate two classification scenarios: (i) models trained on unquantized features and tested on quantized features, and (ii) models trained and tested on quantized features using the same bit depth. This analysis allows us to quantify the trade-off between feature precision, storage efficiency, and classification accuracy.

7.3. Device Type Classification

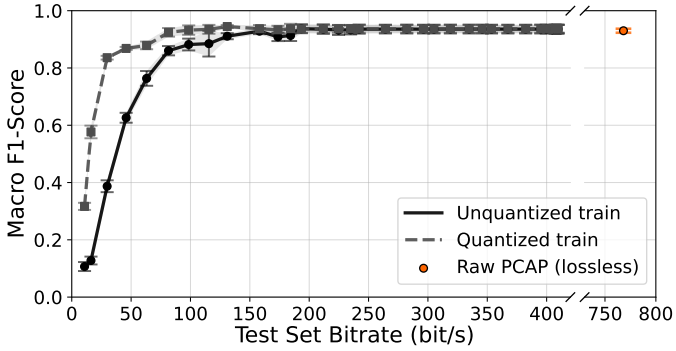
Figure 7a reports the impact of lossy feature compression on device type classification performance, showing the macro F1-score as a function of the average storage bitrate. Results are shown for classifiers trained on unquantized features and evaluated on quantized features, as well as for classifiers trained and tested on quantized representations. For reference, the performance achieved using lossless compression of raw `.pcap` traces is also indicated.

For both lossy configurations, classification performance improves with increasing bitrate, following a saturating trend. At very low bitrates, aggressive quantization leads to a marked degradation in performance, whereas moderate increases in bitrate rapidly restore classification accuracy.

When the classifier is trained and evaluated on quantized features, macro F1-scores above 0.9 are achieved at bitrates around 100 bit/s, and near-lossless performance is reached at approximately 150–200 bit/s. Beyond this point, further



(a)



(b)

Figure 7: Impact of training set bit-rate on Macro F1 score for (a) device type and (b) individual device identification classification tasks. Results compare two approaches: models trained and tested on quantized data versus models trained on unquantized data but tested on quantized data.

increases in bitrate yield only marginal improvements. In contrast, models trained on unquantized features converge more slowly and exhibit higher variability at intermediate bitrates.

7.4. Individual Device Identification

Overall, the performance trends observed for individual device identification are similar to those obtained for device type classification. When the classifier is trained and evaluated on quantized features, performance improves rapidly as the bitrate increases. Macro F1-scores above 0.85 are achieved at bitrates around 80–100 bit/s, while near-lossless performance (macro F1 $\approx$ 0.92–0.94) is reached at approximately 150–200 bit/s. Beyond this range, further increases in bitrate yield marginal performance gains, indicating that most device-specific discriminative information is preserved once a moderate feature precision is retained.

In contrast, models trained on unquantized features and evaluated on quantized representations exhibit slower convergence and higher variability at intermediate bitrates, consistently underperforming the quantized-training configuration. This behavior further confirms that exposing the classifier to quantization effects during training improves robustness to reduced feature precision.

7.5. Discussion and Implications for IoT Forensics

The results of this section show that storage strategies for Zigbee traffic impact not only classification accuracy, but also the set of forensic and analytical tasks that can be supported. Lossless storage of raw packet traces preserves complete information and enables the widest range of forensic analyses, including retrospective feature extraction and protocol-level inspection. However, even with state-of-the-art compression, this approach requires storage rates on the order of several hundred bit/s. Feature-based storage provides a more compact representation at the cost of analytical flexibility. When lossy compression through feature quantization is adopted, storage requirements can be reduced by approximately 4–5$\times$ compared to lossless raw traffic storage, while maintaining near-lossless classification accuracy for both device type classification and individual device identification. Training classifiers directly on quantized features further improves robustness to reduced precision and enables the most storage-efficient operating points. These gains come with functional limitations. Feature-based representations restrict subsequent analyses to the predefined feature set, preventing packet-level inspection and the extraction of alternative features. As a result, lossy feature compression is well suited for continuous monitoring and classification tasks at the edge, whereas lossless raw traffic storage remains preferable for scenarios requiring full forensic flexibility.

8. Conclusions

In this work, we introduced ZIOTP2026, a publicly available multi-topology dataset of Zigbee network traffic collected from 21 commercial smart home devices. With over 252,000 IEEE 802.15.4 frames captured under realistic interaction scenarios, the dataset enables systematic investigation of topology-dependent effects that are largely overlooked in existing IoT traffic datasets.

Using ZIOTP2026, we showed that state-of-the-art traffic classification methods achieve excellent performance under intra-topology evaluation, but suffer significant degradation when evaluated across different network topologies, particularly for fine-grained device identification. These results highlight network topology as a key factor limiting the generalizability of Zigbee traffic analysis models.

We also analyzed the trade-off between traffic storage requirements and classification accuracy. Our results show that lossy feature compression through quantization can reduce storage requirements by approximately 4–5× compared to lossless raw traffic storage, while preserving near-lossless classification accuracy. Overall, this work emphasizes the need for topology-aware analysis and storage-efficient designs in scalable IoT forensic systems.

Future work will focus on extending the dataset to additional topologies and device ecosystems, and on developing topology-aware analysis and adaptive storage strategies for scalable IoT forensic systems.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT in order to assess grammar and rephrase sentences to increase overall text clarity. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

References

[1] U. Cisco, Cisco annual internet report (2018–2023) white paper, Cisco: San Jose, CA, USA 10 (1) (2020) 1–35.

[2] M. Stoyanova, Y. Nikoloudakis, S. Panagiotakis, E. Pallis, E. K. Markakis, A Survey on the Internet of Things (IoT) Forensics: Challenges, Approaches, and Open Issues, IEEE Comm. Surveys & Tutorials 22 (2) (2020) 1191–1221. doi:10.1109/COMST.2019.2962586.

[3] F. Palmese, A. E. C. Redondi, M. Cesana, Designing a forensic-ready wi-fi access point for the internet of things, IEEE Internet of Things Journal 10 (23) (2023) 20686–20702. doi:10.1109/JIOT.2023.3304423.

[4] F. Palmese, A. E. C. Redondi, M. Cesana, Feature-sniffer: Enabling iot forensics in openwrt based wi-fi access points, in: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT), 2022, pp. 1–6. doi:10.1109/WF-IoT54382.2022.10152146.

[5] F. Palmese, A. E. C. Redondi, Collecting channel state information in wi-fi access points for iot forensics, in: 2023 21st Mediterranean Communication and Computer Networking Conference (MedComNet), 2023, pp. 176–183. doi:10.1109/MedComNet58619.2023.10168865.

[6] Ieee standard for local and metropolitan area networks–part 15.4: Low-rate wireless personal area networks (lr-wpans), IEEE Std 802.15.4-2011 (Revision of IEEE Std 802.15.4-2006) (2011) 1–314doi:10.1109/IEEESTD.2011.6012487.

[7] F. De Keersmaeker, Y. Cao, G. K. Ndonda, R. Sadre, A survey of public iot datasets for network security research, IEEE Communications Surveys & Tutorials 25 (3) (2023) 1808–1840.

[8] D.-G. Akestoridis, M. Harishankar, M. Weber, P. Tague, Crawdad cmu/zigbee-smarthome (2022). doi:10.15783/c7-nvc6-4q28.
URL https://dx.doi.org/10.15783/c7-nvc6-4q28

[9] F. Sadikin, T. Van Deursen, S. Kumar, A zigbee intrusion detection system for iot using secure and efficient data collection, Internet of Things 12 (2020) 100306.

[10] T. J. Bihl, K. W. Bauer, M. A. Temple, Feature selection for rf fingerprinting with multiple discriminant analysis and using zigbee device emissions, IEEE transactions on information forensics and security 11 (8) (2016) 1862–1874.

[11] H. J. Patel, M. A. Temple, R. O. Baldwin, Improving zigbee device network authentication using ensemble decision tree classifiers with radio frequency distinct native attribute fingerprinting, IEEE transactions on reliability 64 (1) (2014) 221–233.

[12] S. Dadkhah, H. Mahdikhani, P. K. Danso, A. Zohourian, K. A. Truong, A. A. Ghorbani, Towards the development of a realistic multidimensional iot profiling dataset, in: 2022 19th Annual International Conference on Privacy, Security and Trust (PST), 2022, pp. 1–11. doi:10.1109/PST55820.2022.9851966.

[13] O. Lourme, M. Hauspie, Zigbee dataset (2023). doi:10.57745/NDW74U.
URL https://doi.org/10.57745/NDW74U

[14] N. Keleşoğlu, Ł. Sobczak, Zigbeenet: Decrypted zigbee iot network traffic dataset in smart home environment, Applied Sciences 14 (23) (2024). doi:10.3390/app142310844.
URL https://www.mdpi.com/2076-3417/14/23/10844

[15] R. Li, W. Zhang, L. Wu, Y. Tang, X. Xie, K. Lakshmanna, Zpa: A smart home privacy analysis system based on zigbee encrypted traffic, Wirel. Commun. Mob. Comput. 2023 (jan 2023). doi:10.1155/2023/6731783.
URL https://doi.org/10.1155/2023/6731783

[16] L. Babun, H. Aksu, L. Ryan, K. Akkaya, E. S. Bentley, A. S. Uluagac, Z-iot: Passive device-class fingerprinting of zigbee and z-wave iot devices, in: ICC 2020 - 2020 IEEE International Conference on Communications (ICC), 2020, pp. 1–7. doi:10.1109/ICC40277.2020.9149285.

[17] A. Acar, H. Fereidooni, T. Abera, A. K. Sikder, M. Miettinen, H. Aksu, M. Conti, A. Sadeghi, A. S. Uluagac, Peek-a-boo: I see your smart home activities, even encrypted!, CoRR abs/1808.02741 (2018). arXiv:1808.02741.
URL http://arxiv.org/abs/1808.02741

[18] A. Boiano, A. E. Cesare Redondi, M. Cesana, Iotscent: Enhancing forensic capabilities in internet of things gateways, in: 2023 IEEE 9th World Forum on Internet of Things (WF-IoT), 2023, pp. 1–6. doi:10.1109/WF-IoT58464.2023.10539461.

[19] D.-G. Akestoridis, M. Harishankar, M. Weber, P. Tague, Zigator: analyzing the security of zigbee-enabled smart homes, in: Proceedings of the 13th ACM Conference on Security and Privacy in Wireless and Mobile Networks, WiSec '20, Association for Computing Machinery, New York, NY, USA, 2020, p. 77–88. doi:10.1145/3395351.3399363.
URL https://doi.org/10.1145/3395351.3399363

[20] X. Guo, J. Quan, J. Hou, H. Zhou, X. He, T. He, Accurately identify and localize commodity devices from encrypted smart home traffic, in: 2022 18th International Conference on Mobility, Sensing and Networking (MSN), 2022, pp. 663–670. doi:10.1109/MSN57253.2022.00109.

[21] O. Lourme, G. Grimaud, M. Hauspie, Zbds2023: A multi location zigbee dataset to build innovative iot intrusion detection systems, in: 2023 19th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), 2023, pp. 84–91. doi:10.1109/WiMob58348.2023.10187745.

[22] H. Patel, Non-parametric feature generation for rf-fingerprinting on zigbee devices, in: 2015 IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA), 2015, pp. 1–5. doi:10.1109/CISDA.2015.7208645.

[23] T. J. Bihl, K. W. Bauer, M. A. Temple, Feature selection for rf fingerprinting with multiple discriminant analysis and using zigbee device emissions, IEEE Transactions on Information Forensics and Security 11 (8) (2016) 1862–1874. doi:10.1109/TIFS.2016.2561902.

[24] C. Dubendorfer, B. Ramsey, M. Temple, Zigbee device verification for securing industrial control and building automation systems, in: J. Butts, S. Shenoi (Eds.), Critical Infrastructure Protection VII, Springer Berlin Heidelberg, Berlin, Heidelberg, 2013, pp. 47–62.

[25] F. Palmese, A. E. Redondi, M. Cesana, Resource optimization for evidence collection and preservation in iot forensics-ready access points, IEEE Transactions on Network and Service Management (2025).

[26] H. Assistant, Home assistant operating system (2024).
URL https://www.home-assistant.io/os/